

main

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|  <b>yqc01</b> update readme ✓ | 38ac237 · 3 months ago |  |
|  assets                       | init                   | 3 months ago                                                                        |
|  train                        | init                   | 3 months ago                                                                        |
|  .gitignore                   | add gitignore          | 3 months ago                                                                        |
|  LICENSE                      | init                   | 3 months ago                                                                        |
|  README.md                  | update readme          | 3 months ago                                                                        |
|  README_CN.md               | update readme          | 3 months ago                                                                        |

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## Model Introduction

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**Hy3 preview** is a 295B-parameter Mixture-of-Experts (MoE) model with 21B active parameters and 3.8B MTP layer parameters, developed by the Tencent Hy Team. Hy3 preview is the first model trained on our rebuilt infrastructure, and the strongest we've shipped so far. It improves significantly on complex reasoning, instruction following, context learning, coding, and agent tasks.

| Property                               | Value                              |
|----------------------------------------|------------------------------------|
| Architecture                           | Mixture-of-Experts (MoE)           |
| Total Parameters                       | 295B                               |
| Activated Parameters                   | 21B                                |
| MTP Layer Parameters                   | 3.8B                               |
| Number of Layers (excluding MTP layer) | 80                                 |
| Number of MTP Layers                   | 1                                  |
| Attention Heads                        | 64 (GQA, 8 KV heads, head dim 128) |
| Hidden Size                            | 4096                               |
| Intermediate Size                      | 13312                              |

|                      |                              |
|----------------------|------------------------------|
| Context Length       | 256K                         |
| Vocabulary Size      | 120832                       |
| Number of Experts    | 192 experts, top-8 activated |
| Supported Precisions | BF16                         |

## Highlights

- **STEM & Reasoning** — Complex reasoning underpins everything else. Hy3 preview performs well on challenging STEM benchmarks like FrontierScience-Olympiad and IMOAnswerBench, and achieved excellent results in the Tsinghua Qiuzhen College Math PhD qualifying exam (Spring '26) and the China High School Biology Olympiad (CHSBO 2025), demonstrating generalizable reasoning capacity.
- **Context Learning & Instruction Following** — Real-world tasks require the ability to parse messy, lengthy contexts and follow complex rules. We built CL-bench and CL-bench-Life from our own business scenarios to innovatively measure context learning ability. Hy3 preview exhibits solid gains in both context learning and instruction following capabilities.
- **Code & Agent** — Coding and agents saw the biggest gains. With a rebuilt RL infrastructure and larger-scale training tasks, we posted competitive scores across mainstream coding agent benchmarks (SWE-bench Verified, Terminal-Bench 2.0) and search agent benchmarks (BrowseComp, WideSearch).

## Benchmark Results

### Pre-trained Model Performance

| Category | Benchmark (Metric) | # Shots | Kimi-K2 BASE | DeepSeek-V3 BASE | GLM-4.5 BASE | p |
|----------|--------------------|---------|--------------|------------------|--------------|---|
|          | #ActivatedParams   | -       | 32B          | 37B              | 32B          | 2 |
|          | #TotalParams       | -       | 1043B        | 671B             | 355B         | 2 |
| English  | MMLU               | 5-shot  | 88.24        | 87.68            | 87.73        | 8 |
|          | MMLU-Pro           | 5-shot  | 65.98        | 63.98            | 63.67        | 6 |

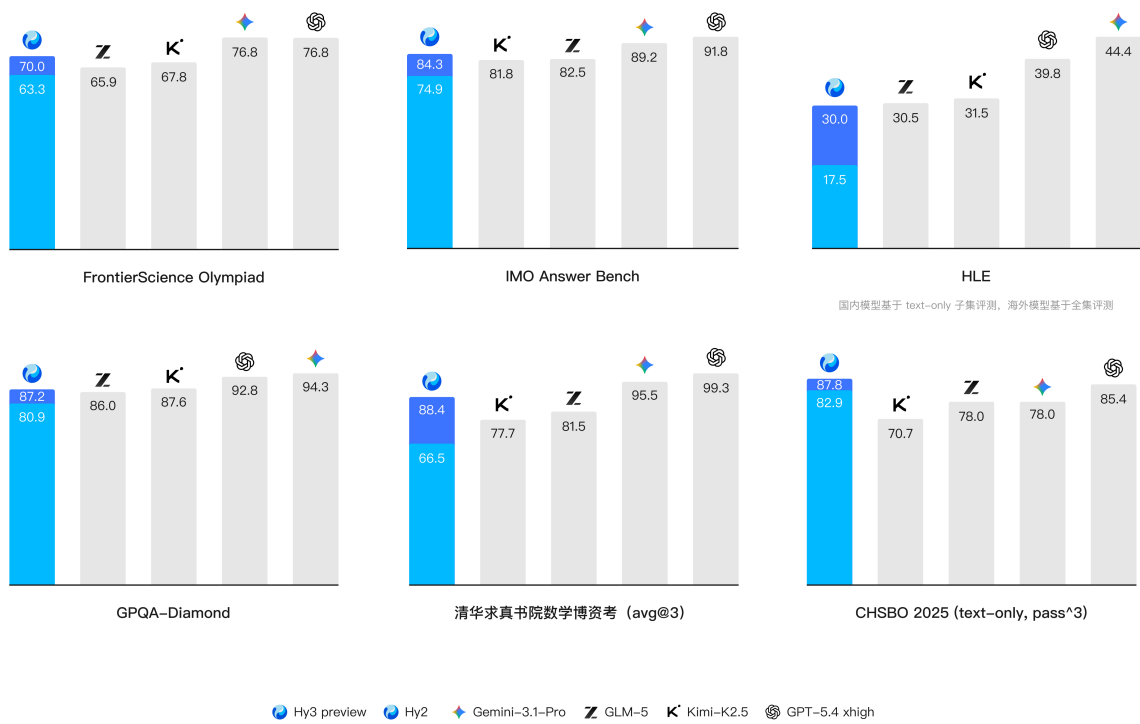
|                |                  |        |              |              |       |   |
|----------------|------------------|--------|--------------|--------------|-------|---|
|                | MMLU-Redux       | 5-shot | <b>87.18</b> | 86.81        | 86.56 | 8 |
|                | ARC-Challenge    | 0-shot | <b>96.66</b> | 94.65        | 96.32 | 9 |
|                | DROP             | 5-shot | 86.40        | <b>86.50</b> | 82.90 | 8 |
|                | PIQA             | 4-shot | <b>84.93</b> | 84.22        | 84.71 | 8 |
|                | SuperGPQA        | 5-shot | 51.10        | 46.17        | 49.64 | 5 |
|                | SimpleQA         | 5-shot | <b>34.37</b> | 26.15        | 29.26 | 2 |
| <b>Code</b>    | MBPP-plus        | 3-shot | <b>81.35</b> | 75.47        | 78.05 | 7 |
|                | CRUXEval-I       | 3-shot | 68.01        | 67.79        | 68.51 | 7 |
|                | CRUXEval-O       | 3-shot | 69.62        | <b>71.00</b> | 67.75 | 6 |
|                | LiveCodeBench-v6 | 1-shot | 30.86        | 29.31        | 27.43 | 3 |
| <b>Math</b>    | GSM8K            | 4-shot | 93.46        | 88.15        | 90.06 | 9 |
|                | MATH             | 4-shot | 71.20        | 59.37        | 61.00 | 7 |
|                | CMath            | 4-shot | 90.83        | 85.50        | 89.33 | 9 |
| <b>Chinese</b> | C-Eval           | 5-shot | <b>91.51</b> | 90.35        | 85.84 | 8 |
|                | CMMLU            | 5-shot | <b>90.72</b> | 87.90        | 86.46 | 8 |
|                | Chinese-simpleQA | 5-shot | <b>74.58</b> | 68.72        | 68.49 | 6 |

|              |         |        |       |       |       |   |
|--------------|---------|--------|-------|-------|-------|---|
| Multilingual | MMMLU   | 5-shot | 77.63 | 79.54 | 79.26 | 8 |
|              | INCLUDE | 5-shot | 75.66 | 77.86 | 76.27 | 7 |

## Instruct Model Performance

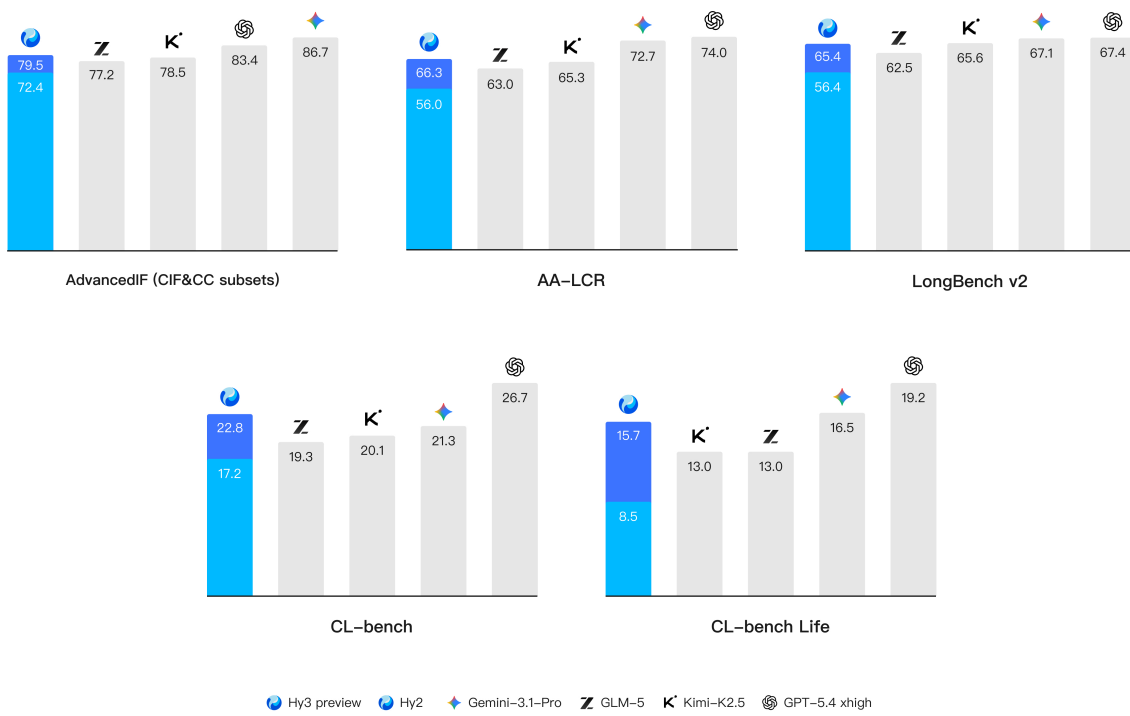
### STEM & Reasoning

Complex reasoning underpins everything else. Hy3 preview performs well on challenging STEM benchmarks like FrontierScience-Olympiad and IMOAnswerBench. It also achieved excellent results in the Tsinghua Qiuzhen College Math PhD qualifying exam (Spring '26) and the China High School Biology Olympiad (CHSBO 2025), demonstrating a high degree of generalizable reasoning capacity.



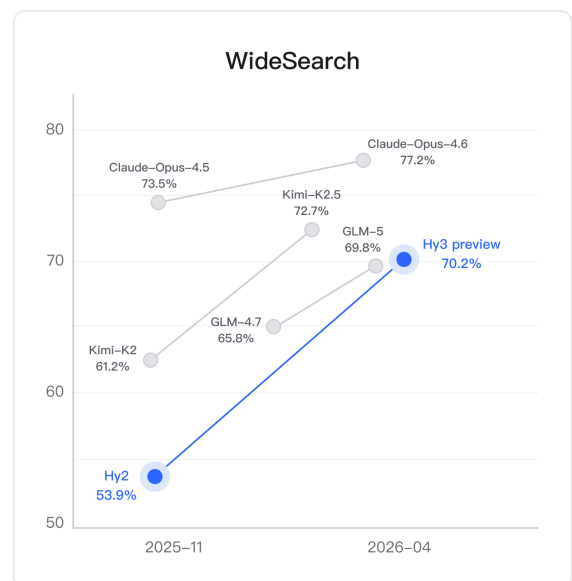
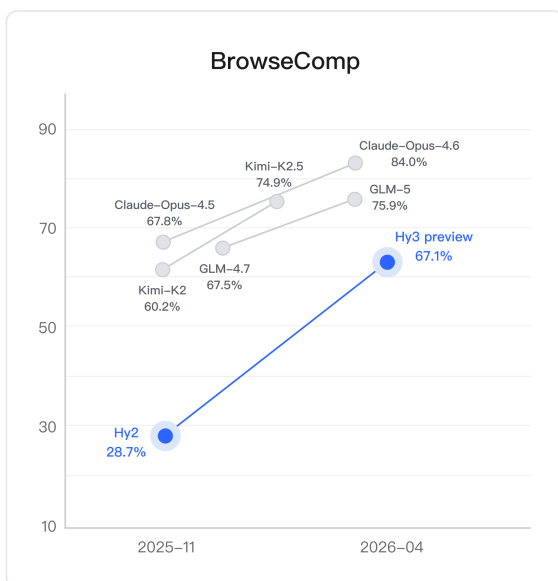
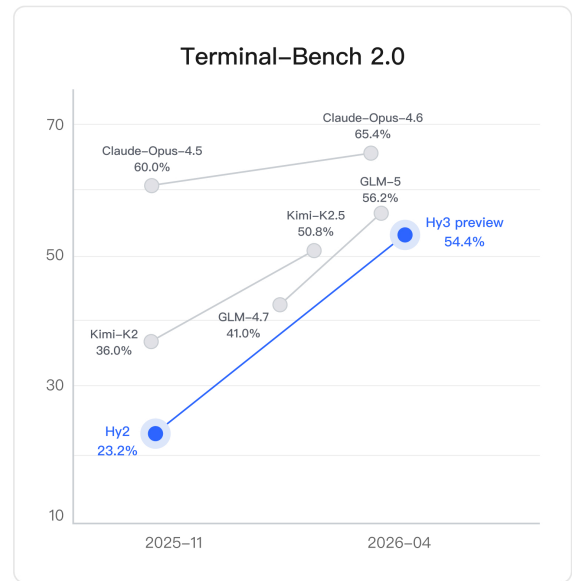
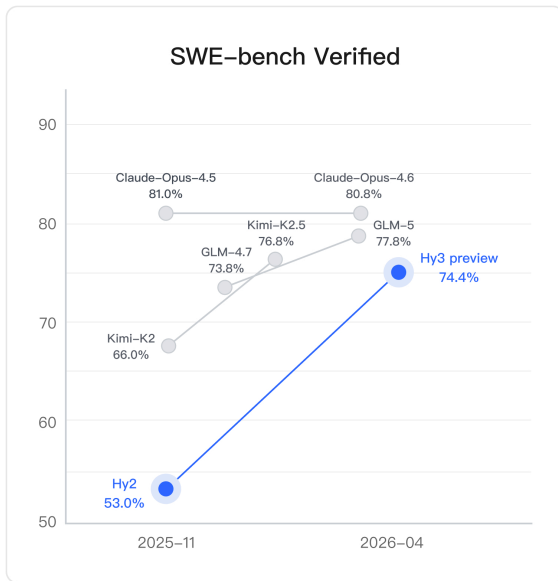
### Context Learning & Instruction Following

Real-world tasks require the ability to parse messy, lengthy contexts and follow complex rules. We built CL-bench and CL-bench-Life from our own business scenarios to innovatively measure context learning ability. Hy3 preview exhibits solid gains in both context learning and instruction following capabilities.

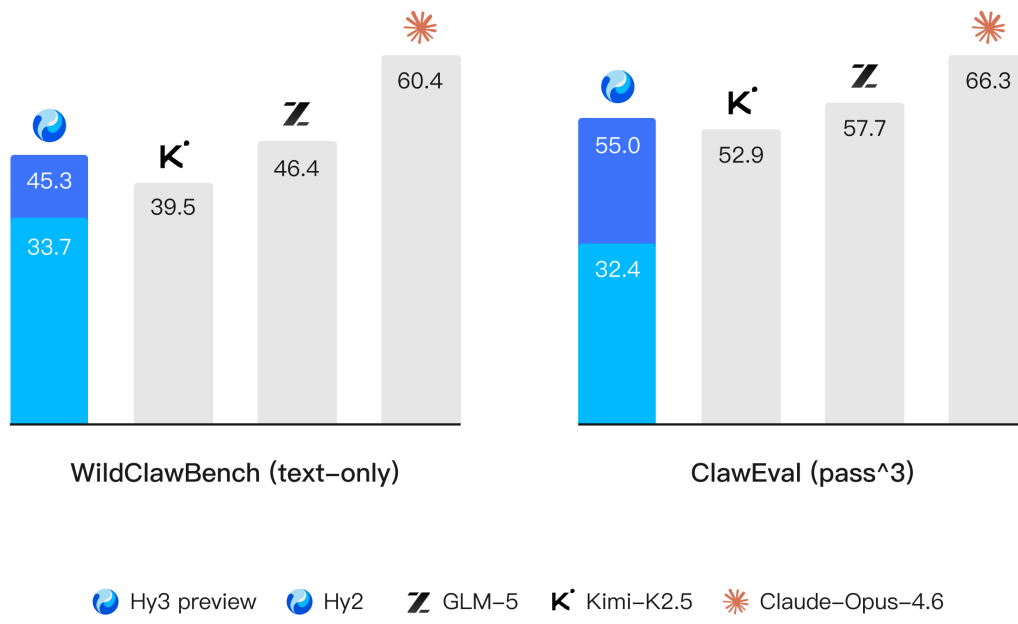


## Code & Agent

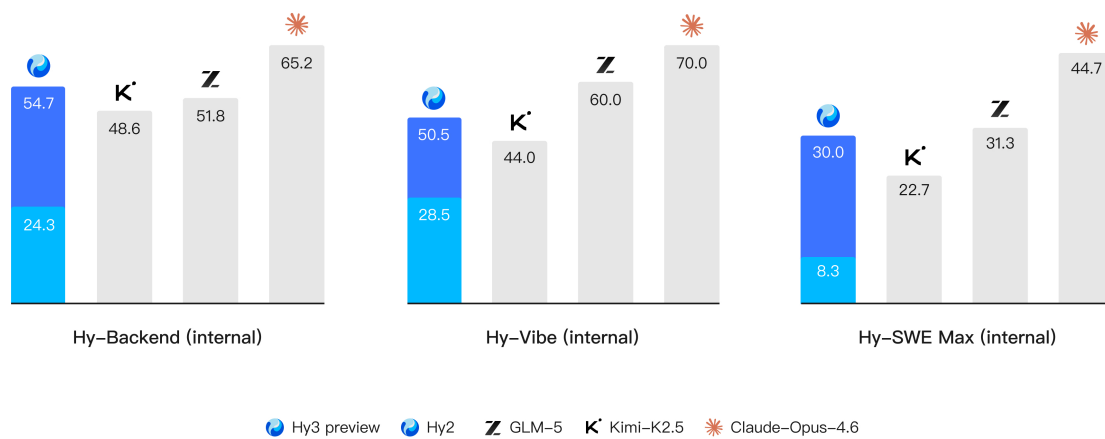
Coding and agents saw the biggest gains. With a rebuilt RL infrastructure and larger-scale training tasks, we posted competitive scores across mainstream coding agent benchmarks (SWE-bench Verified, Terminal-Bench 2.0) and search agent benchmarks (BrowseComp, WideSearch).



Coding is about whether a model can execute in a development environment. Search is about whether it can find and combine information from the open web. Both matter for complex agent scenarios like OpenClaw. Hy3 preview scores well on ClawEval and WildClawBench — a sign that its agent capabilities are becoming practical.



Beyond public benchmarks, we built internal evaluation sets to test the model in real development scenarios. On Hy-Backend (backend-focused tasks), Hy-Vibe Bench (real-user dev workflows), and Hy-SWE Max, Hy3 preview scores competitively against other open-source models.



## News

- [2026-04-23] 🔥 We open-source **Hy3 preview** model weights on [Hugging Face](#), [ModelScope](#), and [GitCode](#).

## Model Links

| Model Name       | Description            | Hugging Face                                                                                            | ModelScope            | GitCode               |
|------------------|------------------------|---------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|
| Hy3 preview      | Instruct model         |  <a href="#">Model</a> | <a href="#">Model</a> | <a href="#">Model</a> |
| Hy3 preview-Base | Pre-trained base model |  <a href="#">Model</a> | <a href="#">Model</a> | <a href="#">Model</a> |

## Quickstart

Deploy Hy3 preview with [vLLM](#) or [SGLang](#) first, then call the OpenAI-compatible API:

```
from openai import OpenAI

client = OpenAI(base_url="http://localhost:8000/v1", api_key="EMPTY")

response = client.chat.completions.create(
    model="hy3-preview",
    messages=[
        {"role": "user", "content": "Hello! Can you briefly introduce"},
    ],
    temperature=0.9,
    top_p=1.0,
    # reasoning_effort: "no_think" (default, direct response), "low",
    extra_body={"chat_template_kwargs": {"reasoning_effort": "no_thin
})
print(response.choices[0].message.content)
```

**Recommended parameters:** `temperature=0.9`, `top_p=1.0`.

**Reasoning mode:** Set `reasoning_effort` to `"high"` for complex tasks (math, coding, reasoning) or `"no_think"` for direct responses.

See the [Deployment](#) section below for how to start the API server.

## Deployment

Hy3-preview has 295B parameters in total. To serve it on 8 GPUs, we recommend using H20-3e or other GPUs with larger memory capacity.

### vLLM

Build vLLM from source:

```
uv venv --python 3.12 --seed --managed-python
source .venv/bin/activate
git clone https://github.com/vllm-project/vllm.git
cd vllm
uv pip install --editable . --torch-backend=auto
```



Start the vLLM server with MTP enabled:

```
vllm serve tencent/Hy3-preview \
  --tensor-parallel-size 8 \
  --speculative-config.method mtp \
  --speculative-config.num_speculative_tokens 1 \
  --tool-call-parser hy_v3 \
  --reasoning-parser hy_v3 \
  --enable-auto-tool-choice \
  --served-model-name hy3-preview
```



## SGLang

Build SGLang from source:

```
git clone https://github.com/sgl-project/sglang
cd sglang
pip3 install pip --upgrade
pip3 install "transformers>=5.6.0"
pip3 install -e "python"
```



Launch SGLang server with MTP enabled:

```
python3 -m sglang.launch_server \
  --model tencent/Hy3-preview \
  --tp 8 \
```



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## Releases

No releases published

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## Packages

No packages published

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Contributors 1

 yqc01

Languages

